Tomáš Šoltinský

Curriculum Vitae

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Employment

2024-26 **Postdoctoral Research Grant Holder**, *National Institute of Astrophysics - Astronomical Observatory of Trieste*, Trieste, Italy

2023-24 **Postdoctoral Research Fellow**, *Tata Institute of Fundamental Research*, Mumbai, India
Department of Theoretical Physics

Education

2019-23 **PhD in Physics**, *The University of Nottingham*, Nottingham, UK
Supervisors: Dr James S Bolton, Dr Nina Hatch
Thesis: Simulating neutral hydrogen across cosmic time: the 21-cm forest in late reionization models and the global 21-cm signal at cosmic dawn
- Vice-Chancellor's Scholarship for Research Excellence
- Graduated July 2023

2014-19 **MPhys in Physics, Astrophysics**, *The University of Edinburgh*, Edinburgh, UK
- Graduated July 2019 with the First Class

2016-17 **Exchange program**, *California Institute of Technology*, Pasadena, USA
- GPA: 3.3

Observing Proposals

2026 **co-I of uGMRT Cycle 51 observations (ID: 51_117)**, *Mapping the Uncharted Forest: 21-cm Absorption Against High-Redshift Sources*, $10\,\mathrm{hr}$ awarded

2026 **PI of uGMRT Cycle 50 observations (ID: 50_010)**, *A uGMRT Band-2 Search for the 21-cm Forest 1D Power Spectrum at* $z \simeq 5.8$, $60\,\mathrm{hr}$ awarded

Software

2024 **21cm-forest_1DPS repository**, https://github.com/tomassoltinsky/21cm-forest_1DPS
Forward-modelling the 21-cm forest with instrumental features and compute its 1D power spectrum
Used in 3 publications

Technical and Academic Experience

2025 **Refereeing scientific papers**, *for A&A, PRL, PRD*

2023-24 **In-Kind Directable Software Development for the LSST-DESC**, *Tata Institute of Fundamental Research*, Mumbai, India
- Development of a code for calculating 3D trispectrum from N-body simulations

2018-19 **MPhys Project**, *The University of Edinburgh*, Edinburgh, UK
Extreme Variability of Active Galactic Nuclei (report) - Supervisor: Prof Andy Lawrence
- Tested black hole accretion disc models in the context of extreme variability of AGN luminosity
- Included a public summary (created a card game)

2018 **Cormack Vacation Research Scholarship and Deans' Vacation Scholarship**, *The Institute for Astronomy*, Edinburgh, UK
21cm Signal Marking the End of Dark Ages (report, poster) - Supervisor: Dr José Oñorbe
- Mock observations of global 21cm signal with focus on Dark Ages based on Nyx simulations

2018 **Senior Honours Project**, *The University of Edinburgh*, Edinburgh, UK
Scalar Field Dark Energy - Supervisor: Prof Andy Taylor
- Simulated the expansion of the universe driven by dark energy modeled as a dynamical scalar field

2017-18 **Telescope Group Project**, *The University of Edinburgh*, Edinburgh, UK
Measurement of the Characterising Properties of Exoplanet TrES-2b and its Host Star - Supervisor: Prof Philip Best
- Transit Photometry and Doppler Spectroscopy methods

2017 **Summer Undergraduate Research Fellowship**, *Caltech*, Pasadena, USA

Redefining the Habitable Zone - Supervisor: Prof David J. Stevenson

- Determining conditions under which water in liquid phase can be present on the surface of a planet

2017 **Research in Experimental Physics**, *Caltech, LIGO*, Pasadena, USA

Signal Processing - Supervisor: Dr Thomas J. Massinger

- Data analysis and investigation of glitches in data

Publications

T Šoltinský, G Autieri, V Iršič, et al., "The impact of the IGM thermal state on the Lyα flux 3D power spectrum from linear to highly non-linear scales", submitted to *JCAP*, 2026. (link)

G Autieri, V Iršič, **T Šoltinský**, et al., "Physical Calibration of a Minimal Effective Field Theory of the Three-Dimensional Lyman-α Forest", submitted to *JCAP*, 2026. (link)

T Šoltinský, A Chakraborty, G Kulkarni, et al., "Forest without Trees is still Fruitful: Constraints on the thermal state of the neutral IGM at $z \approx 5.6$ with the 21-cm forest power spectrum", submitted to *Nature Communications*, 2026. (link)

CD Science Working Group, "Overview of 21cm Experiments at high redshift with SKAO", published in Advancing Astrophysics with the SKAII (AASKAII), 2026 (arXiv:2606.20366). (link)

J Cang, B Ciardi, B Maity, et al., "Exploring the Cosmic Dawn through the 21 cm Forest and High-redshift Radio Sources with the SKA", published in Advancing Astrophysics with the SKAII (AASKAII), 2026 (arXiv:2606.20366). (link)

S K Patil, **T Šoltinský**, S Maitra, et al., "Efficient neutral-IGM inference from noisy 21-cm forest spectra with latent-space U-Net encoding and XGBoost", published in *MNRAS*, Volume 546, Issue 4, id.stag236, 18 pp., 2026. (link)

T Šoltinský, G Kulkarni, S P Tendulkar, et al., "Prospects of a statistical detection of the 21-cm forest and its potential to constrain the thermal state of the neutral IGM during reionization", published in *MNRAS*, Volume 537, Issue 1, pp.364-378, 2025. (link)

E Puchwein, J S Bolton, L C Keating, et al. "The Sherwood-Relics simulations: overview and impact of patchy reionization and pressure smoothing on the intergalactic medium", published in *MNRAS*, Volume 519, Issue 4, pp.6162-6183, 2023. (link)

T Šoltinský, J S Bolton, M Molaro, et al. "Probing quasar lifetimes with proximate 21-centimetre absorption in the diffuse intergalactic medium at redshifts $z \geq 6$", published in *MNRAS*, Volume 519, Issue 2, pp.3027-3045, 2023. (link)

T Šoltinský, J S Bolton, N Hatch, et al. "The detectability of strong 21-cm forest absorbers from the diffuse intergalactic medium in late reionization models", published in *MNRAS*, Volume 506, pp. 5818-5835, 2021. (link)

C S Cockell, "Astrobiology: Understanding Life in the Universe", *Wiley-Blackwell*, 2015

- Contributed to and was cited in the Preface to the book

Awards, Scholarship, Grants

2019-22 **Vice Chancellor's Scholarship for Research Excellence (EU)**, *The University of Nottingham*, Nottingham, UK

To support my PhD studies

2019 **Students into the World grant**, *The Tatra banka Foundation*, Slovakia

2018 **RSE Cormack Undergraduate Astronomy Research Prize**, *Royal Society of Edinburgh*, Edinburgh, UK

For the "21cm Marking the End of Dark Ages" project

2016 **Year 2 Physics Class medal**, *The University of Edinburgh*, Edinburgh, UK

For excellent performance during the 2015/16 session

2014 **The memorial letter of St. Gorazd**, *The Ministry of Education, Science, Research and Sport of the Slovak Republic*, Bratislava, Slovakia

For successful representation of the Slovak Republic at Austrian Young Physicists' Tournament and achievement of bronze medal

2014 **Best in Natural Sciences school award**, *Pavol Horov Gymnasium*, Michalovce, Slovakia

For representation of the school at various Physics and Maths competitions

Mentoring and Teaching

2021 **Mentoring at GROWNi.sk**, Slovakia

- Advising students in their career and study decisions

2016 **Outreach talks**, *Pavol Horov Gymnasium*, Michalovce, Slovakia

- "From LA through half the globe to simulated universe", Topic: career path and simulations in astrophysics
- "From LA to LA", Topic: gained experience and how to study abroad
- "Science has no borders", Topic: my research experience and ongoing research in the world

2023 **Advanced 21-cm Cosmology School and Workshop**, *National Institute of Science Education and Research*, Bhubaneswar, India, Lectures (link)

Service to Academic Community

2023-24 **State of the Universe Seminar Series Organizer**, *TIFR*, Mumbai, India

2022 **Astronomy Postgraduate Student Representative**, *The University of Nottingham*, Nottingham, UK

- Main point of communication between PG students and staff

2021-22 **Lunch Talks Organizer**, *The University of Nottingham*, Nottingham, UK

2019-22 **Physics Coursework Marker**, *The University of Nottingham*, Nottingham, UK

2017-21 **Student Ambassador**, *The University of Edinburgh and the University of Nottingham*, Edinburgh and Nottingham, UK

- Resided over the information desk and gave tours of the campus and the School of Physics and Astronomy at the Open Days, Post-offer Visit Days, Visiting Students Welcome Event

2018 **UUKi Parliamentary Reception for the Go International: Stand Out Campaign**, *Westminster*, London, UK

- Ambassador for The University of Edinburgh

Presentations at International Conferences and Meetings

2026 **Tatra Astro Summit 2026**, *Astronomical Institute of the Slovak Academy of Sciences*, Stará Lesná, Slovakia, Talk

2026 **Cosmology at Home 2026**, Recorded Talk and Chaired a session (link)

2026 **SKA CD/EoR Science Team Meeting 2026**, *Palace of Sciences*, Belgrade, Serbia, Talk

2026 **EAS 2026**, *Swisstech convention centre*, Lausanne, Switzerland, Talks and posters at 3 sessions (link 1, link 2)

2026 **Cosmology Seminar**, *The Institut d'Astrophysique Spatiale*, Orsay-Paris, France, Seminar

2026 **Cosmology Journal Club**, *University of Padova*, Padova, Italy, Seminar

2025 **The Fifth National Workshop on the SKA Project - From precursors to SKAO: shaping the future of Italian radio astronomy**, *CNR*, Bologna, Italy, Talk

2025 **Kaba Kada: Exploring the first billion years of the Universe**, *Oaks Port Douglas Resort*, Port Douglas, Queensland, Invited talk and part of panel

2025 **ICRAR-Curtin Colloquium**, *Curtin University*, Perth, Western Australia, Colloquium

2025 **NAM 2025**, *Durham University*, Durham, UK, Talk and a poster at 2 sessions

2025 **The galaxy-IGM connection in the first billion years**, *IFPU*, Trieste, Italy, Talk

2025 **Radio Cosmology and Continuum Observations in the SKA Era: A Synergic View**, *International Centre for Theoretical Sciences*, Bengaluru, India, Talk (link)

2024 **SKA Cosmology SWG Meeting**, *Observatoire de la Cote d'Azur*, Nice, France, Talk

2024 **IFPU colloquium**, *IFPU*, Trieste, Italy, Colloquium

2024 **The First Gigayear(s)**, *Hilo Hawaiian Hotel*, Hilo, Hawai'i, Talk

2024 **The Origin and Evolution of Supermassive Black Holes**, *Sexten Center for Astrophysics*, Sexten, Italy, Talk (link 1, link 2)

2024 **Cosmic Dawn at High Latitudes Conference**, *Royal Swedish Academy of Sciences*, Stockholm, Sweden, Talk (link)

2024 **State of the Universe**, *TIFR*, Mumbai, India, Seminar (link)

2024 **Cosmology in the Alps Conference**, Les Diablerets, Switzerland, Talk (link)

2024 **Monday Afternoon Talks**, *MIT Kavli Institute*, Cambridge, US, Talk

2024 **Pune-Mumbai Cosmology and Astro-Particle meeting**, *The Inter-University Centre for Astronomy and Astrophysics*, Pune, India, Talk and Chaired a session (link)

2024 **Astronomical Society of India meeting**, *IISc*, Bengaluru, India, Talk

2024 **HEA group meeting**, *Masaryk University*, Brno, Czechia, Seminar

2023 **Astrophysics Seminar**, *University of Helsinki*, Helsinki, Finland, Seminar

2023 **Galaxy coffee**, *Max Planck Institute for Astronomy*, Heidelberg, Germany, Talk

2023 **21-cm Series Seminar**, *University of Cambridge*, Cambridge, UK, Seminar

2023 **Colloquium**, *National Centre for Radio Astrophysics*, Pune, India, Colloquium

2022 **Lunch talk**, *The University of Nottingham*, Nottingham, UK, Talk

2022 **21-cm Series Seminar**, *University of Cambridge*, Cambridge, UK, Seminar

2021 **State of the Universe**, *TIFR*, Mumbai, India, Seminar (link)

2021 **NAM 2021**, *The University of Bath*, Bath, UK, Talk

2021 **SAZERAC 2.0**, Recorded talk (link)

2021 **Lunch talk**, *The University of Nottingham*, Nottingham, UK, Talk

2020 **Lunch talk**, *The University of Nottingham*, Nottingham, UK, Talk

Membership in collaborations

2024 **SKA Cosmology and Epoch of Reionization SWGs**, link Cosmology, link EoR

2024 **21-cm Cosmology Group (India)**, link

2019 **Sherwood-Relics simulation suite**, link

2023-24 **LSST-DESC, Large scale structure WG**, provisional membership, link

Training/Skills

2023 **Foundation HPC-Skills Course**, *DiRAC*

2023 **DP0 Virtual Summer School 2023**, *Vera C. Rubin Observatory/LSST*

Programming Python, Jupyter notebook, C, LaTeX, Mathematica, IDL, Radiative Transfer, 21cmFAST, IRAF, SExtractor, Large data analysis, MCMC, OpenMP parallelisation, Slurm